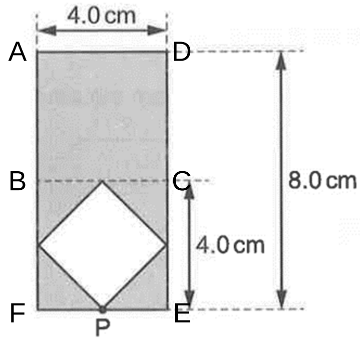
		$u_1 + u_2 = v_2 - v_1$
5	**B**	Method 1 (by calculation):  Taking moments about P, (ignore thickness since metal sheet is uniform) Clockwise moments of ADEF = Clockwise moments of ABCD + Clockwise moments of BCEF weight of ADEF $\times d =$ weight of ABCD $\times$ 6 cm $+$ weight of BCEF $\times$ 2 cm $(4\times 4\times g + 2\times 4\times g)d = (4\times 4\times g\times 6) + (2\times 4\times g\times 2)$ $d = 4.667$ ≈ 4.7 cm Method 2 (by observation and intuition):

<table>
<tr><th>Qn</th><th>Answer</th><th>Solution</th></tr>
<tr><td></td><td></td><td>

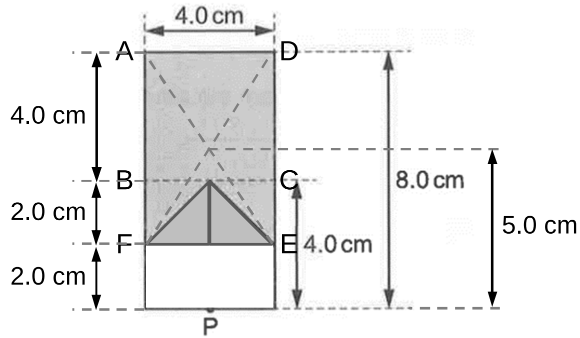


Fig. 5.1

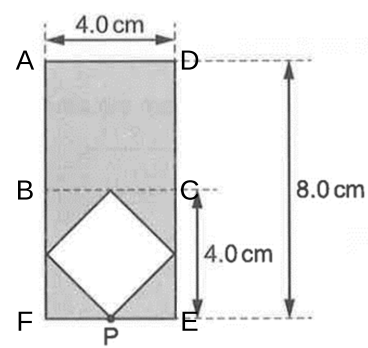


Fig. 5.2

In Fig. 5.1, the centre of gravity of sheet of metal ADEF is at a distance of 5.0 cm from point P.

By observation, the centre of gravity of sheet of metal ADEF in Fig. 5.2 must be at a shorter distance from point P, compared to that in Fig. 5.1. Hence, 4.7 cm is a logical choice.

</td></tr>
<tr><td>6</td><td>B</td><td>Output power of wind turbine generator</td></tr>
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